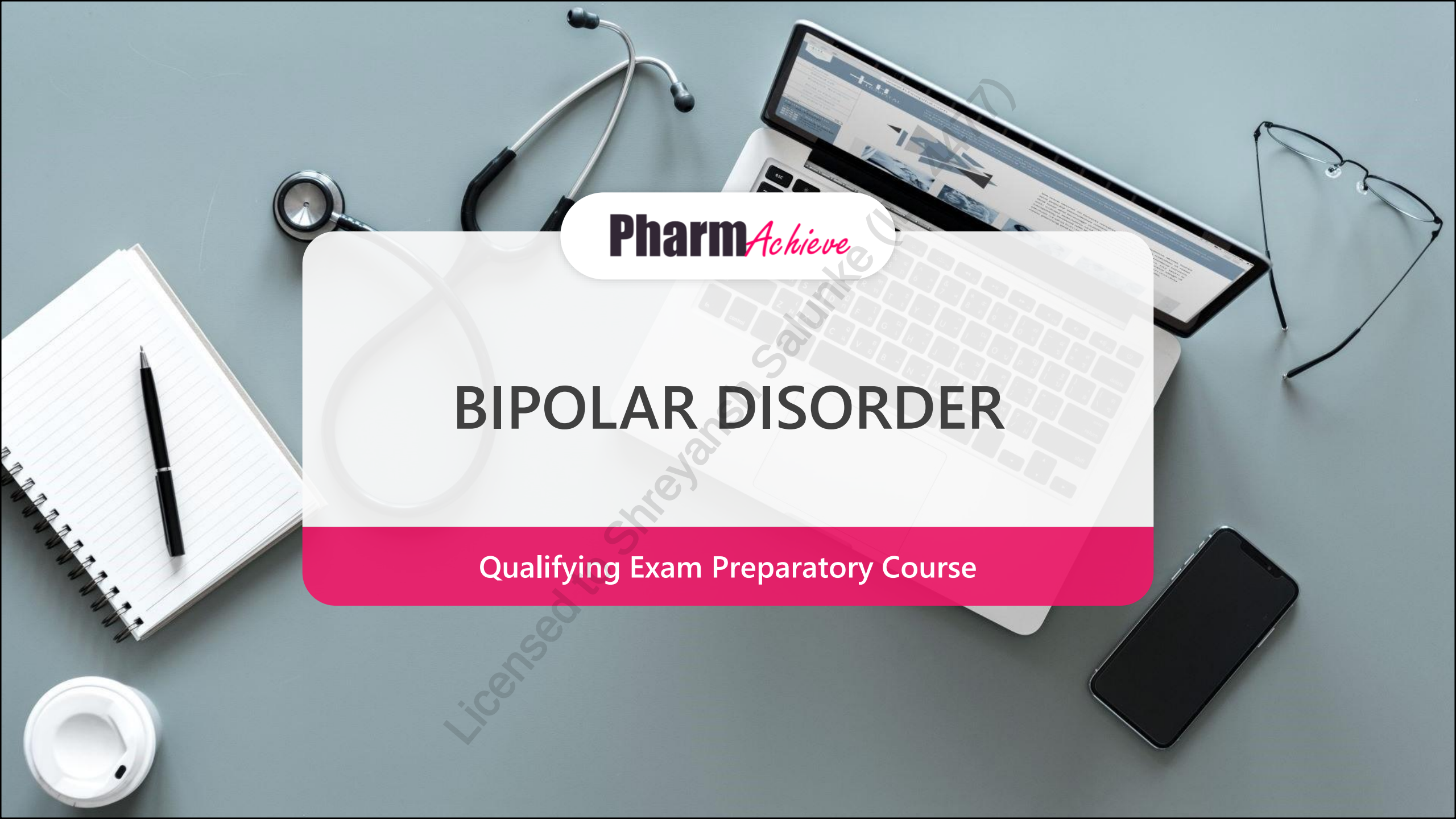


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

BIPOLAR DISORDER

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AAP Atypical Antipsychotic

ACEI Angiotensin-Converting Enzyme Inhibitors

ACTH Adrenocorticotrophic Hormone

ADR Adverse Drug Reaction

BAD I Bipolar Disorder Type I

BPD Bipolar Disorder

CBT Cognitive Behavioural Therapy

CNS Central Nervous System

DVP Divalproex

ECG Electrocardiogram

ECT Electroconvulsive Therapy

EPS Extrapyrimal Symptoms

HIV Human Immunodeficiency Viruses

IM Intramuscular

LAI Long-Acting Injection

Li Lithium

MAOI Monoamine Oxidase Inhibitor

MDQ Mood Disorder Questionnaire

NSAIDs Non-Steroidal Anti-Inflammatory Drugs

ODT Orally disintegrating tablet

PO By Mouth

SE Side Effect

SJS Stevens Johnson Syndrome

SNRI Serotonin and Norepinephrine Reuptake Inhibitor

SSRI Selective Serotonin Reuptake Inhibitor

TEN Toxic Epidermal Necrolysis

TCA Tricyclic Antidepressant

DEFINITION

Bipolar Disorder is a chronic mental illness with **cyclic** fluctuations in mood (mania, hypomania, and depression), energy, and behaviour

Bipolar Type I

At least **1 manic episode** +/- **major depressive episode** or **hypomania**

Bipolar Type II

At least **1 major depressive episode** + at least 1 episode of **hypomania** + **no history of manic episode**

Other Types

- Unspecified BPD
- Cyclothymic Disorder
- Substance/medication induced BPD
- BPD due to another medical condition
- Other specified BPD

Pathophysiology of bipolar disorder is **unknown**, many hypotheses exist

RISK FACTORS AND EPIDEMIOLOGY

Risk Factors

- Substance abuse
- Family history of bipolar disorder
- Psychosocial or physical stress
- Head trauma
- Perinatal insult



Epidemiology

- Adolescents / young adults (15-30 years)
 - The average age of onset for bipolar disorder is ~ 25 years
 - 1st episode in females is more likely to be a major depressive episode
 - Females are more likely to get mixed states, depressive episodes and rapid cycling
 - 1st episode in males is more likely to be a manic episode

CLINICAL PRESENTATION

Manic Episode

- **Elevated self-esteem**
- **↓ need for sleep**
- More and faster speech
- Racing thoughts or easily distracted
- More goal-directed activity, or psychomotor agitation
- Impulsive engagement in activities that have high potential for painful consequences
- **Mood disturbance is severe enough to cause a marked impairment in social or occupational functioning/ psychotic features**
- Episode is not attributable to the effects of a substance or another medical condition
- **Symptoms present for ≥ 1 week and present nearly every day**

Hypomanic Episode

Similar to mania except:

- Associated with an unequivocal change in functioning that is uncharacteristic
- **Episode is not severe enough** to cause marked impairment in social or occupational function or to necessitate hospitalization
- **No psychotic features** present
- **Lasting at least 4 consecutive days and present most of each day**

CLINICAL PRESENTATION

Major Depressive Episode

- **Depressed mood most of the day** as reported by self/others (children & adolescents may be irritable)
- Markedly diminished interest/pleasure in all/almost all, activities most of the day, nearly every day
- Significant weight loss or gain, or decrease or increase in appetite nearly every day
- Insomnia or hypersomnia nearly every day
- Psychomotor agitation or retardation nearly every day that others can observe
- Fatigue or loss of energy nearly every day
- Feelings of worthlessness or excessive, inappropriate, or delusional guilt nearly every day
- Diminished ability to think or concentrate, or indecisiveness, nearly every day
- **Recurrent thoughts of suicidal ideation** without a specific plan or a suicide attempt or a specific plan for committing suicide
- **Symptoms present for > 2 weeks**

OTHER CAUSES OF MANIC SYMPTOMS/DIFFERENTIAL DIAGNOSES

Medical Causes	
CNS disorders	Brain tumor, strokes, head injuries, multiple sclerosis, systemic lupus erythematosus, etc.
Infections	Encephalitis, sepsis, HIV
Electrolyte or metabolic abnormalities	Calcium or sodium fluctuations, hyper or hypoglycemia
Endocrine or hormonal dysregulation	Cushing's disease, hyper or hypothyroidism, menstrual-related or pregnancy-related or premenopausal mood disorders

Drug Causes	
Drug withdrawal states	• Benzodiazepines • Antidepressants • CNS stimulants • Alcohol and illicit drug intoxications
Drug use	• Anabolic steroids • ACTH/cortisol • Corticosteroids • Caffeine / Theophylline • Ephedra • Pseudoephedrine • Thyroid preparations
Herbal products	• St. John's Wort

GOALS OF THERAPY

- ✓ Control symptoms of acute episode
- ✓ Prevent recurrences/relapse by ensuring patient continues therapy lifelong
- ✓ Provide supportive care for comorbid psychiatric conditions (e.g. anxiety) and medical conditions (e.g. diabetes)
- ✓ Provide relevant education (e.g. psychoeducation) to allow greater insight for psychiatric/medical conditions

NON-PHARMACOLOGICAL THERAPY

ADJUNCTS TO MEDICATIONS



Discontinue: Antidepressants, caffeine, nicotine and stimulants in MANIA; sedative-hypnotics / benzos in DEPRESSION (if possible); treat substance abuse



Electroconvulsive Therapy (ECT): For severe/non-responsive symptoms or if pregnant (**2nd line**): 3-8 sessions, but medications are usually still required in the maintenance phase; generates a faster effect in patients compared to medications



Group Psychoeducation Programs: Educate patients, develop coping strategies and learn to recognize episodes; more effective in mania (**1st line for maintenance phase**)



Cognitive Behavioural Therapy: More effective in depressive bipolar disorder (**2nd line for both maintenance and acute depression**)



Ensure Safety of Patients and Others: Assess for psychotic features & suicidal ideation; treat substance abuse

No evidence for psychosocial interventions in acute mania

PHARMACOLOGICAL THERAPY

Mood Stabilizers:
takes 2-4 weeks to show efficacy

Lithium Carbonate – the most important medication in BPD

- First-line in acute mania, maintenance, and mild depression
- Available as IR and CR
- Measure lithium levels 12 hours after dose at least every 3-6 months once levels have stabilized
 - Acute mania levels: 0.8-1.2 mmol/L
 - Maintenance levels: 0.6-1.0 mmol/L
 - Lower targets in elderly: 0.4-0.8mmol/L
- Monitor renal and thyroid function at least every 6 months; check electrolytes periodically
- May cause hyperparathyroidism, thyroid abnormalities (hypothyroidism more common than hyperthyroidism), hypercalcemia, impaired neurocognition, ataxia
- Avoid dehydration and large changes in sodium consumption, as this can affect lithium levels
Caffeine can decrease lithium levels. NSAIDs and ACEIs can increase lithium levels
- Patients who experience tremors on lithium therapy may benefit from:
 - Decrease or eliminate caffeine
 - Reduce lithium dose
 - Add beta-blocker (propranolol/atenolol)
 - Avoid during lactation

PHARMACOLOGICAL THERAPY

Mood Stabilizers:
takes 2-4 weeks to show efficacy

Valproic Acid/Divalproex

- Therapeutic drug monitoring is more common during the initiation and titration phases
- Caution in women of childbearing age – **teratogenic**
- SE: weight gain, thrombocytopenia, hepatotoxicity (rare), polycystic ovary syndrome
- Divalproex: Compound of valproic acid and sodium valproate which breaks down into valproic acid in gastrointestinal tract. Causes less stomach upset than valproic acid due to enteric coated formulation

Lamotrigine

- Must be titrated slowly to avoid severe rash (Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis)
- SE: Weight neutral, significant nausea
- DDI with divalproex (lamotrigine levels are doubled with divalproex)

Carbamazepine

- Effective in mixed states
- Must be titrated slowly to avoid severe rash (Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis)
- Therapeutic monitoring to avoid toxicity. No relationship between serum level and efficacy

PHARMACOLOGICAL THERAPY

Atypical Antipsychotics

Risperidone

- Benztropine can be used to reduce EPS
- Available as long-acting injectable

Paliperidone

- Active metabolite of risperidone (similar SE profile)
- Renal dose adjustment required
- Minimal drug interactions due to lack of liver metabolism

Olanzapine

- Caution with metabolic syndrome
- IM 1st line in agitation

Asenapine

- No activity at muscarinic receptors in the therapeutic dose range; therefore, it **does not cause any anticholinergic ADRs**
- Should not eat or drink for a minimum of 10 minutes after administration (food/drink may reduce bioavailability)
- May have an unpleasant taste and may cause tongue numbness/tingling

PHARMACOLOGICAL THERAPY

Atypical Antipsychotics

Aripiprazole

- **IM 1st line in agitation**
- Available as long-acting injectable

Ziprasidone

- Twice daily dosing
- Must be **given with food** (minimum **500 calories**)
- **Weight gain neutral**
- Baseline ECG should be done

Cariprazine

- Effective for both **mania** (at higher dose) and **depression** (at lower dose)

Quetiapine

- **Caution in metabolic syndrome**
- **Can impair neurocognition**
- **Metabolized by CYP 3A4**

Lurasidone

- Must be **given with food**
- **Weight gain neutral**

Lumateperone

FYI – as not available on the Canadian market yet)

- Common adverse effects are nausea and fatigue
- Low incidence of **EPS** and an increase in **prolactin**

PHARMACOLOGICAL THERAPY

Acute Mania: Choose regimens in the order presented, unless patient factors specify otherwise

First Line Monotherapies

- Lithium
- Quetiapine
- Divalproex/Valproic Acid
- Asenapine
- Aripiprazole
- Paliperidone >6mg
- Risperidone
- Cariprazine

First Line Combination Therapies*

- Quetiapine + Li/DVP
- Aripiprazole + Li/DVP
- Risperidone + Li/DVP
- Asenapine + Li/DVP

*Combination therapy is **more effective** and has a faster onset

Second Line Therapies

- | | |
|-----------------------|---------------|
| • Olanzapine | • Ziprasidone |
| • Carbamazepine | • Haloperidol |
| • Olanzapine + Li/DVP | • ECT |
| • Lithium + DVP | |

Better evidence for efficacy and tolerability/safety

If symptoms are not controlled with first-line agents, dosing should be optimized.

If no improvement in symptoms within the first 2 weeks at therapeutic doses, factors should be considered before making changes to medications (e.g. potential nonadherence and concurrent substance use)

If treatment with optimal dose of 1st line monotherapy or combo is inadequate/intolerable, switch to or add on an alternate 1st line treatment

PHARMACOLOGICAL THERAPY

Acute Depression: Choose regimens in the order presented, unless patient factors specify otherwise

First Line Therapies

- Quetiapine
- Lurasidone + Li/DVP
- Lithium
- Lamotrigine
- **Cariprazine (new, with increased clinical experience)**
- Adjunctive lurasidone

Second Line Therapies

- Divalproex/Valproic acid
- Adjunctive SSRI/Bupropion
- ECT
- Olanzapine + fluoxetine
- Lumateperone (new evidence suggests may have role in bipolar I and II depression)



Better evidence for efficacy and tolerability/safety

If lack of response to the first trial: consider switching to or adding on an alternate 1st line agent

- Switching is preferred over add-on to limit the degree of polypharmacy (*unless* there was a partial response)

Patients (and families, as appropriate) should be advised of the risk of treatment-emergent affective switch

- Antidepressants should be discontinued if early warning signs of mania manifest
- Antidepressants should *not* be used as monotherapy for the treatment of bipolar I depression (can ↑ risk of inducing mania)

PHARMACOLOGICAL THERAPY

Maintenance

- Prevent relapse and restore functioning and quality of life
- Medications that have been effective in the acute phase should be continued during the maintenance phase (except for chronic use of antidepressants, which is generally not recommended in bipolar disorder)
- Ongoing psychoeducation and flexible, collaborative engagement with patients are recommended to optimize the acceptability of maintenance pharmacological treatment

Maintenance: Prevention of mood episode

First Line Therapies

- Lithium
- Quetiapine
- Divalproex
- Lamotrigine
- Asenapine
- Quetiapine + Li/DVP
- *Aripiprazole + Li/DVP
- *Aripiprazole PO
- *Aripiprazole LAI

Second Line Combination Therapies

- Olanzapine
- Risperidone LAI
- Adjunctive Risperidone LAI
- Carbamazepine
- Paliperidone > 6mg
- Lurasidone + Li/DVP
- *Ziprasidone + Li/DVP

*Better evidence
for efficacy and
tolerability/safety*

Note: multiple 1st
line options
should be trialed
before moving on
to 2nd line options

*Aripiprazole/Ziprasidone has **not** been shown to prevent depression, only mania

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
*Symptoms: Moods, thoughts, behaviour, sleep * Use <i>Mood Disorder Questionnaire (MDQ)</i>	Improved	2-4 weeks
Safety Endpoints		
Parameter	Degree of Change	Timeframe
SJS/TEN	None	Up to 8 weeks post-initiation
Weight gain	Manageable/tolerable	Duration of therapy
Nausea/vomiting	Manageable/tolerable	First few weeks of therapy
Extrapyramidal symptoms	None	Duration of therapy
Metabolic Syndrome (A1C, blood glucose)	Manageable	Duration of therapy

Duration of therapy varies based on patient specific factors such as symptoms, pregnancy, adverse effects, etc. Some patients require lifelong therapy

PREGNANCY AND BREASTFEEDING

- **Conventional antipsychotics and risperidone increase serum prolactin levels**, interfering with ovulation and decreasing fertility; thus, such agents may require discontinuation to increase the likelihood of conception
- **Medication-free pregnancy:**
 - Gradual taper before conception
 - Ensure the patient has been clinically stable for at least 4-6 months and is considered at low risk of relapse
- **Avoid valproate products** (valproic acid, divalproex sodium) in female children and those of childbearing age
- Discussion needs to be held between the patient and the healthcare provider on the best options during pregnancy and lactation

TAKEAWAY/SUMMARY OF THERAPY RECOMMENDATIONS

1. Try 1st line agents at optimal dosages for at least 2-4 weeks before switching/adding on other 1st line agents
2. First-line monotherapies for acute mania: lithium, quetiapine, DVP, asenapine, aripiprazole, paliperidone > 6mg, risperidone, or cariprazine
 - Combination therapy can also be used as first line
3. First-line monotherapies for acute depression: quetiapine, lurasidone+ Li/DVP, lithium, lamotrigine, cariprazine, adjunctive lurasidone
4. First-line monotherapies for maintenance therapy: lithium, quetiapine, DVP, lamotrigine, asenapine, quetiapine + Li/DVP, aripiprazole + Li/DVP, aripiprazole PO, aripiprazole LAI

CASE 1

TK is a 20-year-old engineering student who also works part-time at a local print shop. For the past two weeks, he has not been sleeping well because he is studying for exams. Today, he showed up late to work and told his manager he had been racing his car on the highway and had been stopped for speeding. He would not follow his manager's direction and instead decided that he would take apart one of the photocopiers in the shop so that he could "make it work better." TK also told his co-worker that he had been building a "super copier" at home and that he was going to sell it for "billions of dollars." TK's manager and co-worker noted that TK has been speaking more rapidly and loudly over the past week, and he has also been having trouble focusing on his work. TK is normally a quiet and responsible individual, and because this behaviour is very out of character for him, the manager decided to bring TK to the hospital. TK began using cannabis recreationally last year and consumes approximately 0.5 grams 2 to 3 times per week

1. What is TK's diagnosis?

- a) Bipolar disorder I
- b) Bipolar disorder II
- c) Substance-induced mania
- d) Unable to obtain



CASE 1

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2. Which medication should be recommended for TK?

- a) Divalproex
- b) Lithium
- c) Olanzapine
- d) Lithium + divalproex



CASE 1

TK is a 20-year-old engineering student who also works part-time at a local print shop. For the past two weeks, he has not been sleeping well because he is studying for exams. Today, he showed up late to work and told his manager he had been racing his car on the highway and had been stopped for speeding. He would not follow his manager's direction and instead decided that he would take apart one of the photocopiers in the shop so that he could "make it work better." TK also told his co-worker that he had been building a "super copier" at home and that he was going to sell it for "billions of dollars." TK's manager and co-worker noted that TK has been speaking more rapidly and loudly over the past week, and he has also been having trouble focusing on his work. TK is normally a quiet and responsible individual, and because this behaviour is very out of character for him, the manager decided to bring TK to the hospital. TK began using cannabis recreationally last year and consumes approximately 0.5 grams 2 to 3 times per week

3. TK is started on lithium. What does NOT need to be monitored?

- a) Thyroid function
- b) Liver function
- c) Renal function
- d) Serum calcium



CASE 2

TK's manic episode resolved, and he was discharged from the hospital 6 months ago. He has been taking lithium at a dose of 1200 mg daily, and he has tolerated it very well. He does not take any other medications and has no other medical conditions. All his labs are within normal limits, and his lithium level is 0.9 mmol/L. He has decreased his cannabis use (approximately 1-2 times monthly). He comes to the pharmacy reporting a low mood. He recently saw his psychiatrist, who noted that TK is experiencing a depressive episode.

1. What is the most appropriate next step his psychiatrist should take?
 - a) Add lurasidone
 - b) Change to fluoxetine + olanzapine
 - c) Change to bupropion
 - d) Add escitalopram



CASE 2

TK's manic episode resolved, and he was discharged from the hospital 6 months ago. He has been taking lithium at a dose of 1200 mg daily, and he has tolerated it very well. He does not take any other medications and has no other medical conditions. All his labs are within normal limits, and his lithium level is 0.9 mmol/L. He has decreased his cannabis use (approximately 1-2 times monthly). He comes to the pharmacy reporting a low mood. He recently saw his psychiatrist, who noted that TK is experiencing a depressive episode.

2. TK is interested in trying lamotrigine for his depression. Which of the following is the most important consideration?
- a) Pharmacokinetic drug interaction with lithium
 - b) Need for slow dose titration
 - c) Additive weight gain with lithium
 - d) Need for rapid dose titration



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CHANGE LOG

April 2025

- Slide 23: Deleted question 4

May 2025

- Content reviewed; minor grammar changes made

August 2025

- Slide 12: added "available is IR and CR"
- Slide 13: Added information to the drugs

September 2025

- Slide 18: Elaborated on monitoring parameters

November 2025

- Slide 15: Added that lumateperone is not available on Canadian market yet, as it is not on the Drug Product Database but is mentioned in the CANMAT guidelines
- Slide 20: Added detail to summary of recommendations and added 'Takeaway' to title
- Slide 21: Updated Case 1 to add more information about cannabis
- Slide 22: Updated wording of question 2 and answer choices
- Slide 24: Updated question 2

March 2026

- Slide 5: Changed average age of onset
- Slide 12: Added information to valproic acid
- Slide 14: Added FYI to lumateperone since still not available in Canada
- Slides 15,16,17: Added information
- Formatting changes done throughout, content reviewed